PRATIK MANE

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EDUCATION

Indiana University Expected Dec 2026
Master of Science in Applied Data Science
Purdue University Jan 2024 – Dec 2025
Master of Science in Computer Science

PROFESSIONAL EXPERIENCE

Software Engineering Intern, XR4RAP LLC**,** Indianapolis**,** USA June 2025 – Aug 2025

- Built a therapeutic VR pilot for Meta Quest using C#, Unity XR Toolkit, and XR Origin to support immersive guided experiences
- Integrated API-based AI services for narration, music, sound effects, and visual assets while evaluating IP-safe content sources
- Implemented locomotion, teleportation, interactions, and timed scenes; resolved camera and UI conflicts to improve user stability
- Authored technical docs and integration guides for AI service evaluation and Unity pipeline setup to streamline future development

XR Technology Lead and Web Developer, Indiana University School of Dentistry**,** Indianapolis**,** USA Jan 2024 – June 2025

- Developed full-stack architecture for the Extended Reality Clinical Practice Education Lab (XRCPEL), building responsive interfaces with HTML5, CSS3, and JavaScript workflows that supported appointment booking, collaboration requests, and daily operations.
- Built appointment-booking and collaboration-request features that replaced manual lab processes with streamlined digital workflows, improving scheduling, communication, and coordination for faculty and students across the program each day at scale
- Built a MindAR image-tracking experience that anchors an interactive 3D Meta Quest model to the ADEA research poster
- Developed an interactive XR research platform for ADEA 2025, combining a Three.js 3D headset experience, AI-guided research exploration, responsive educational panels, analytics, collaboration workflows, and MindAR poster-based augmented reality

Software Developer, RBK Technologies Limited, Pune, India Sept 2022 - Dec 2023

- Led UX planning and Figma prototyping for RBK's primary website, translating business requirements into responsive page structures, reusable components, intuitive navigation, and organized content flows that set clear brand direction for engineering team signoff
- Built and deployed a responsive website with Next.js, React, and Headless UI, delivering smooth desktop and mobile web experiences while optimizing rendering performance, accessibility, and cross-browser compatibility to support consistent user engagement
- Configured Hostinger deployment and Google Analytics (GA4) tracking for traffic, user-behavior, and content decisions
- Used Git and GitHub for version control, reviews, and releases; reduced page load times **25%** through asset optimization and caching

Full Stack Developer Intern, Maharashtra Knowledge Corporation Limited, Pune, India Mar 2022 - Sept 2022

- Developed Digital University using a Vue.js frontend with MongoDB and MySQL backend architecture, supporting **1,000+** concurrent users during peak registration periods through scalable data services and responsive enrollment workflows
- Built a WebSocket and Node.js feedback system, delivering real-time student-instructor communication with **99.9%** message delivery
- Improved responsive layouts with CSS Grid and Flexbox, reducing bounce rates and improving accessibility
- Partnered with product and design teams in Agile sprints, delivering features ahead of project milestones

SKILLS

Languages: JavaScript, Python, C#, C/C++, SQL, HTML, CSS, R
Frameworks and Libraries: React, Next.js, Node.js, Express, Vue.js, OpenCV, pandas, NumPy
Databases: MySQL, MongoDB
Tools and Platforms: Git, GitHub, Docker, Figma

RESEARCH AND PRESENTATIONS

Extended Reality in Dental Clinical Education | American Dental Education Association (ADEA) Annual Session Apr 2026

- Co-investigated a two-cohort study of **193** IU Dental DDS students, evaluating usability, comfort, and dental-anxiety outcomes across virtual, mixed, and augmented reality experiences designed for clinical education and training within predoctoral dental programs
- Co-authored and presented a poster at the American Dental Education Association Annual Session (ADEA), reporting that **98.6%** of participants rated VR beneficial while **94.6%** found VR relaxing during immersive dental education experiences in the study

ACADEMIC PROJECT EXPERIENCE

Ames Housing Market Analysis: Statistical Modeling for Real Estate Investment Strategy Jan 2026 - May 2026

- Conducted statistical analysis of 2,930 Ames Iowa home sales using hypothesis testing, regression, and time series analysis to identify price drivers and market timing patterns; built a regression model (R^2=0.76) and published insights on an interactive Quarto website

Image Processing Toolkit for Edge Detection and Enhancement Jan 2025 - May 2025

- Developed a Python-based image processing toolkit utilizing OpenCV for advanced edge detection, noise reduction, and image enhancement. Implemented algorithms like Sobel, Canny, and Gaussian Blurring to improve visual clarity in medical image datasets

AI-Based Image Captioning using CNN and BLIP Sept 2024 - Dec 2024

- Built an AI-driven image captioning system leveraging Convolutional Neural Networks (CNN) for feature extraction and BLIP (Bootstrapping Language-Image Pretraining) for natural language caption generation